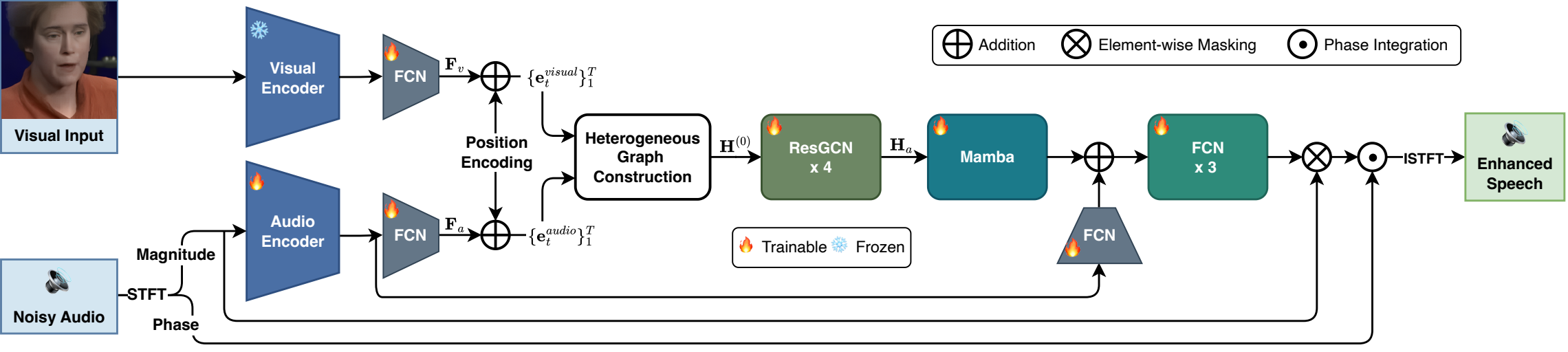
Visual Input
Visual Encoder
FCN
$\mathbf{F}_v$
$\{\mathbf{e}_t^{visual}\}_1^T$
Position Encoding
Heterogeneous Graph Construction
$\mathbf{H}^{(0)}$
ResGCN x 4
$\mathbf{H}_a$
Mamba
FCN x 3
ISTFT
Enhanced Speech
Addition
Element-wise Masking
Phase Integration
Noisy Audio
STFT
Magnitude
Phase
Audio Encoder
FCN
$\mathbf{F}_a$
$\{\mathbf{e}_t^{audio}\}_1^T$
Trainable
Frozen
FCN